

ECE 501 Project Week 1 Report

Project Title: Vehicle Orientation Estimation Using Conventional Image Processing

Group Number: 14

Project number: 8

Team Members:

Het Patel	AU2440221	het.p6@ahduni.edu.in
Shlok Shah	AU2440178	shlok.m2@ahduni.edu.in
Malay Panchal	AU2440050	malay.p@ahduni.edu.in
Dev Prajapati	AU2440022	dev.p12@ahduni.edu.in

Github Repository Link: https://github.com/hetpatel5107/DIP_2026_Project8_Group14

Dataset Overview

For this project, we acquired the DRASHTI-HaOBB dataset. Developed as a first-of-its-kind dataset for unstructured and chaotic Indian traffic scenarios, it features 27,577 high-resolution images captured using a drone.

The primary advantage of this dataset is its use of a Nadir view (a 90-degree straight-downward camera angle). In traditional street-level traffic filming, larger vehicles like buses and trucks completely block smaller vehicles like motorcycles and auto-rickshaws (known as occlusion). A top-down perspective eliminates this problem entirely, giving a clear line of sight to every vehicle on the road.

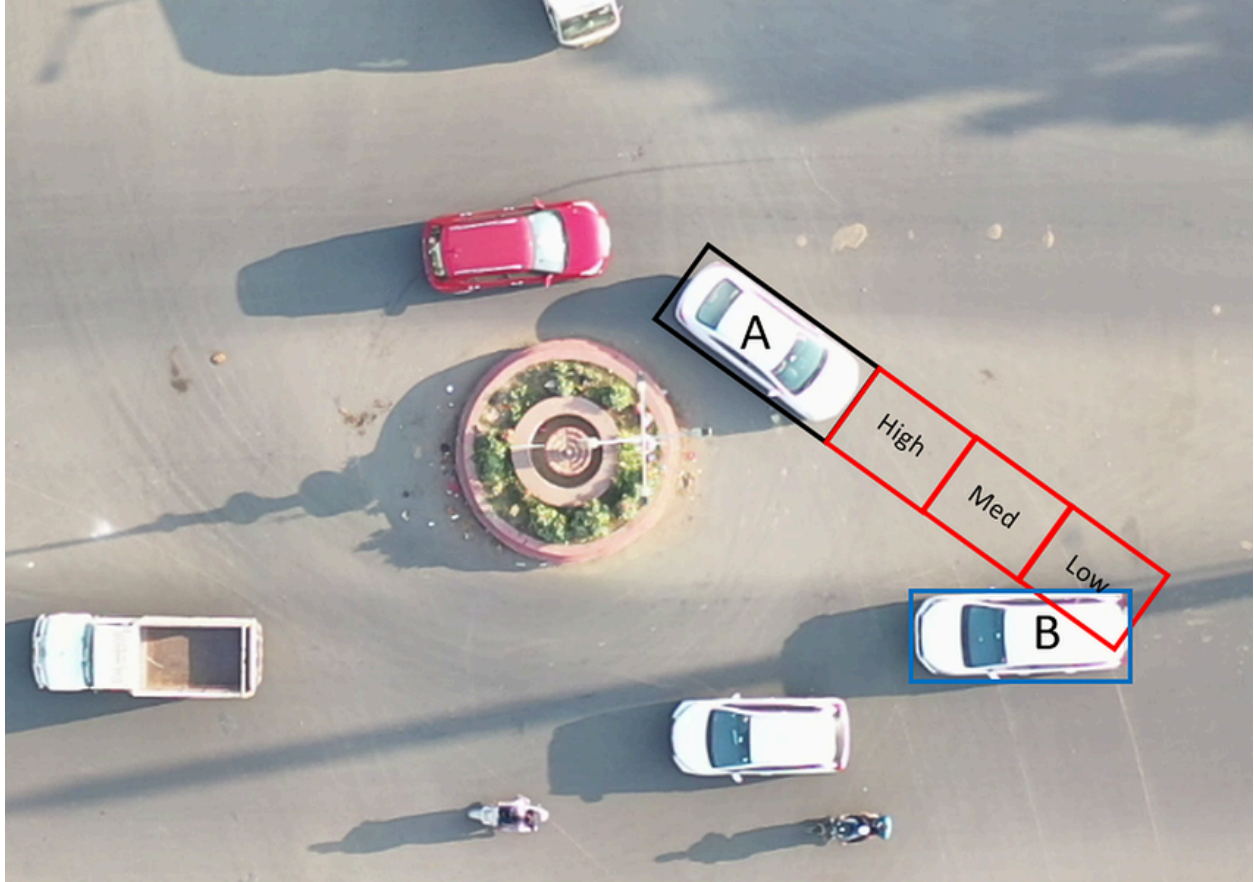


Figure 1: Sample vehicle annotation from the dataset documentation showing Oriented Bounding Boxes (OBB).

Key Information & Fields Explained

After inspecting a small sample file, we found that instead of standard horizontal rectangles, this dataset uses **Oriented Bounding Boxes (OBB)** to tightly wrap around angled or turning vehicles. The relevant data fields for each vehicle are:

1. **Polygon Coordinates (x_1, y_1 to x_4, y_4):** The exact pixel coordinates of the four corners of the vehicle, allowing the bounding box to rotate.
2. **Heading Angle (θ):** A value from 0° to 360° that identifies the absolute direction in which the front of the vehicle is facing. This is the most critical field for our project.
3. **Class ID:** A numeric label identifying the vehicle type across 14 distinct Indian road classes (e.g., AutoRickshaw, Motorcycle, Bus, Truck).
4. **Flight Height:** The altitude of the drone (45 m to 127 m), which allows us to handle scaling differences when vehicles appear smaller or larger.

References

1. **Dataset Paper:** Bhavsar, Y. M., Zaveri, M. S., Raval, M. S., Patel, K. R., & Zaveri, S. B. (2026). "Descriptor: Drone Nadir-View Annotated Images of Vehicles Detection Dataset for India With Heading-Angle Oriented Bounding Box (DRASHTI-HaOBB)." *IEEE Data Descriptions*. Available at [IEEE Xplore](#)
2. **Dataset Repository:** Ahmedabad University Aerial Vision Lab. (2026). *DRASHTI-HaOBB Multi-Class Repository*. Available at [Hugging Face Datasets](#)

Literature Review

Tan and Baker [1] proposed a conventional image-processing approach for vehicle localization and orientation estimation in traffic images. The problem they are working on determines the position, orientation, and class of vehicles from images. Their approach uses image gradient information directly instead of explicit feature extraction and matching. The method also uses assumptions about the ground plane and the generally rectangular shape of road vehicles.

References

[1] T. N. Tan and K. D. Baker, "Efficient image gradient based vehicle localization," in *IEEE Transactions on Image Processing*, vol. 9, no. 8, pp. 1343-1356, Aug. 2000, doi: 10.1109/83.855430.

Link to the reference paper: [Efficient Image Gradient Based Vehicle Localization](#)

Problem Understanding

Our project title is Vehicle Orientation Estimation Using Conventional Image Processing. So, the input to the project is an image containing a vehicle. The next task would be to process the image using conventional image-processing techniques to identify visual features that can indicate the vehicle's orientation. The estimated orientation is then obtained as the output and compared with the real orientation provided by the dataset.

Input → Image Processing → Orientation Estimation → Output (Vehicle Orientation)

Potential Techniques

The following conventional image-processing techniques can potentially be used for vehicle orientation estimation:

- **Edge Detection:** Detects the boundaries and structural edges of a vehicle, which can provide information about its shape and direction.
- **Sobel Operator:** Calculates image gradients and can help determine the dominant direction of edges in the vehicle image.
- **Hough Transform:** Detects prominent lines in the image, which can help identify the main directional structure of a vehicle.

- Contour Detection: Extracts the outline of the vehicle, allowing its shape and orientation to be analyzed.
- Image Gradient and Orientation Histogram: Uses the directions of image gradients to estimate the dominant orientation of the vehicle.

These techniques can be tested on the dataset to determine which approach is most suitable for estimating vehicle orientation.

1. Work done this week

- Explored the DRASHTI-HaOBB dataset.
- Studied one relevant paper.
- Understood the project problem and expected input/output.
- Investigated potential conventional image-processing techniques.

2. Work to be done next week

- Further explore the dataset.
- Load and visualize sample vehicle images.
- Select a small subset of vehicle images for initial testing.
- Start experimenting with suitable image-processing techniques.

3. Literature Review

- T. N. Tan and K. D. Baker, "Efficient image gradient based vehicle localization," in *IEEE Transactions on Image Processing*, vol. 9, no. 8, pp. 1343-1356, Aug. 2000, doi: 10.1109/83.855430.
- Link to the reference paper: [Efficient Image Gradient Based Vehicle Localization](#)

4. Challenges faced

- Understanding the dataset annotations.
- Identifying the relevant data required for our project.
- Understanding the approach used in the selected research paper.